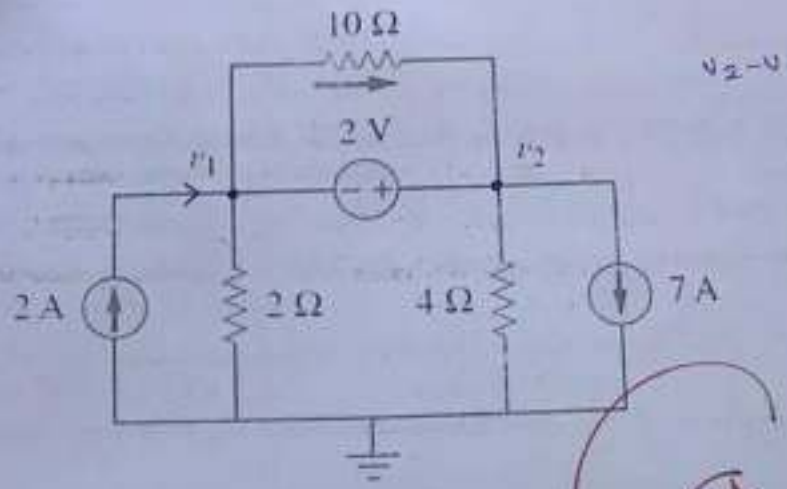
For the questions a, b, c, d & e consider the circuit shown in figure 1 and answer (use nodal method):



$V_2 - V_1 = 2V$

Fig.1

a. The value of V1 is:

$V_1 = -7.33\,V$

b. The value of V2 is:

$V_2 = -5.33\,V$

c. The value of the current (i) through 10Ω resistor is:

$i = 0.2\,A$

c. The power absorbed by 4 Ω resister is:

$P = \frac{V^2}{R} = \frac{(V_2)^2}{4} = \frac{(-5.33)^2}{4} = 7.11\,W$

d. The power absorbed by 2 A current source is:

$P = I V_1 = 2(-5.33) = -10.66\,W$

e. Write the required equations (do not simplify)

$V_a - V_b = 2$ ..... ①

$2 = 7 + \frac{V_1}{2} + \frac{V_2}{4}$ ..... ②

Prob. 2: Consider the circuit shown in figure 2. and answer:

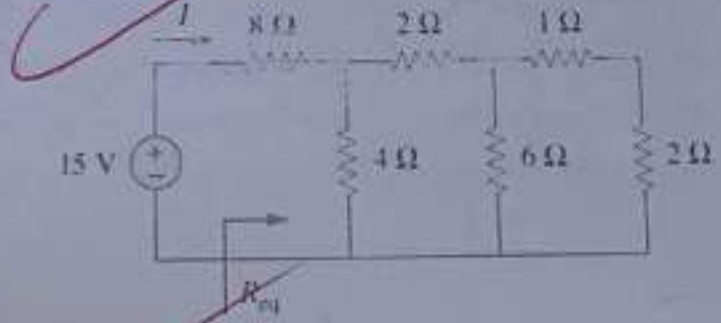


Fig.2

a. The equivalent resistance Req is:

$Req = 10\,\Omega$

b. The current I is:

$I = \frac{\Delta V}{R_{eq}} = \frac{15}{10} = 1.5A$

c. The Power absorbed by 4Ω resistor is :

$P = I^2R = (0.75)^2(4) = 2.25\ W$

Prob.3: a. The charge transferred through an element is given by $q(t)=2t^3+5t^2+6$ C. Calculate the current entering the element at instant t =10 ms.

$i = \frac{dq}{dt} \Rightarrow i = 6t^2 + 10t \Rightarrow i|_{t=10ms} = 6(10\times10^{-3})^2 + 10(10)^{-3} = 600\times10^{-6} + 100\times10^{-3} = 100.6\ mA$

b. A load draws 500W when connected to a 230 V line. Calculate the current and the energy consumed in two hours in joules?

$P = 500W$, $V = 230V$, $t = 2\ hours$

$i = P/V$, $i = \frac{500}{230} \rightarrow 2.17A$

$Energy = Pt$, $= 500\times2\times60\times60$, $\Rightarrow Energy = 3600000$, $= 3.6M\ J$, $= 3.6\times10^6\ J$

Prob.4: Refer to figure 3; write the mesh equations in matrix form:

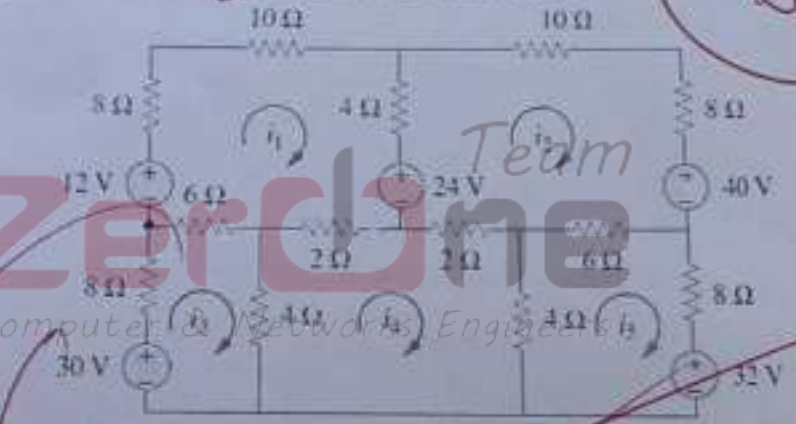


Fig. 3

$$\begin{bmatrix} 30 & -4 & -6 & -2 & 0 \\ -4 & 30 & 0 & -2 & -6 \\ -6 & 0 & 18 & -4 & 0 \\ -2 & -2 & -4 & 8 & 0 \\ 0 & -6 & 0 & -4 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} -12 \\ -16 \\ 30 \\ 0 \\ -32 \end{bmatrix}$$

Good luck